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On the optimality of finding DMDGP symmetries

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Abstract
The Discretizable Molecular Distance Geometry Problem (DMDGP) is a subclass of the Distance Geometry Problem, which aims to embed a weighted simple undirected graph in a Euclidean space, such that the distances between the points correspond to the values given by the weighted edges in the graph. The search space of the DMDGP is combinatorial, based on a total vertex order that implies symmetry properties related to partial reflections around planes defined by the Cartesian coordinates of three immediate and consecutive vertices that precede the so-called symmetry vertices. Since these symmetries allow us to know a priori the cardinality of the solution set and to calculate all the DMDGP solutions, given one of them, it would be relevant to identify these symmetries efficiently. Exploiting mathematical properties of the vertices associated with these symmetries, we present an optimal algorithm that finds such vertices.

Keywords DMDGP · Branch-and-prune · DMDGP symmetries · Protein geometry

Mathematics Subject Classification 51K05 · 68R05 · 05C90

1 The Discretizable Distance Geometry Problem (DMDGP)

The main problem of the Euclidean Distance Geometry (Liberti et al. 2014; Liberti and Lavor 2016; Mucherino et al. 2013), called the Distance Geometry Problem (DGP), is given as follows (recent literature itens about the DGP can be found in Billinge et al. (2016), Billinge et al. (2018), Lavor et al. (2017), Liberti and Lavor (2017), Liberti and Lavor (2018), among others):

Definition 1 Given a simple undirected graph $G = (V, E, d)$ whose edges are weighted by a function $d : E \to [0, \infty)$ and a positive integer K, find a function $x : V \to \mathbb{R}^K$ such that

$$\forall \{u, v\} \in E, \ \ \|x_u - x_v\| = d_{uv}, \tag{1}$$

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where $x_u = x(u)$, $x_v = x(v)$, $d_{uv} = d(\{u, v\})$, and $\|x_u - x_v\|$ is the Euclidean distance between x_u and x_v.

For a general function $d : E \to [0, \infty)$, it may happen that $G = (V, E, d)$ cannot be embedded in $\mathbb{R}^K$, for some positive integer K, satisfying the system (1) (for example, this happens if there is a clique in G such that the related triangular inequality is not satisfied). However, in DGP applications, the function d is related to some existing frameworks and Eq. (1) have a solution. Hence, throughout the paper, we will assume that the DGP solution set is nonempty.

The DGP can be formulated as a global optimization problem,

$$\min_{x_1,\ldots,x_n \in \mathbb{R}^K} \sum_{\{u,v\}\in E} \left(\|x_u - x_v\|^2 - d_{uv}^2\right)^2,$$

where $|V| = n$, but results presented in Lavor et al. (2006) indicate that continuous optimization algorithms do not scale well for medium or large instances. A survey on different methods to the DGP is given in Liberti et al. (2010).

The DMDGP (Lavor et al. 2012a, b) is a combinatorial version of the DGP (Cassioli et al. 2015; Malliavin et al. 2019), first proposed in the context of quantum computing (Lavor et al. 2005; Marquezino et al. 2019) and related to the problem of calculating 3D protein structures ($K = 3$) using distance information provided by Nuclear Magnetic Resonance (NMR) experiments (Crippen and Havel 1988; Lavor et al. 2019; Wüthrich 1989).

The formal definition of the DMDGP (Lavor et al. 2012a) is the following:

Definition 2 Given a DGP graph $G = (V, E, d)$ and a vertex order $v_1, \ldots, v_n$ such that

- v_1, v_2, v_3 can be fixed in $\mathbb{R}^3$ satisfying (1);
- $\forall i > 3$, the set $\{v_{i-3}, v_{i-2}, v_{i-1}, v_i\}$ is a clique with

$$d_{i-3,i-2} + d_{i-2,i-1} > d_{i-3,i-1},$$

find a function $x\colon V \to \mathbb{R}^3$ such that

$$\forall \{u, v\} \in E,\ \|x_u - x_v\| = d_{uv}.$$

The key point in the DMDGP definition is the vertex order (Lavor et al. 2012). For a general DGP graph, DMDGP orders may not exist and the problem to find them is NP-complete (Cassioli et al. 2015). However, for DGP graphs related to protein molecules, a DMDGP order can be obtained a priori (Lavor et al. 2019), based on information provided by protein geometry. Allowing some repetitions in the vertex order, the problem can be solved in polynomial time (Lavor et al. 2019).

The DMDGP order implies symmetry properties related to partial reflections around planes defined by the Cartesian coordinates of three consecutive vertices (see Fig. 1). In Fig. 1, there is a symmetry at vertex v_9, given by the plane defined by the positions of vertices v_6, v_7, v_8, generating two solutions.

Since the DMDGP symmetries allow us to know a priori the cardinality of the solution set and to calculate all the DMDGP solutions, given one of them (Liberti et al. 2013; Mucherino et al. 2012) [(which can be obtained by any algorithm proposed in the literature to solve the DMDGP (Liberti et al. 2014)], it would be relevant to identify the vertices related to these symmetries in an efficient way. This paper presents an optimal algorithm to do that.

The next section characterizes mathematically the DMDGP symmetries and Section 3 describes the main contribution of the paper. Section 4 provides the final comments.

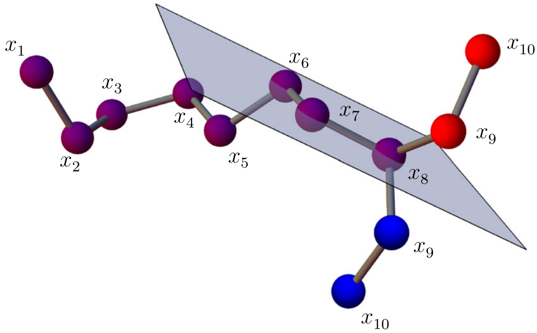

Fig. 1 Two DMDGP solutions with symmetry at vertex v_9

2 BP algorithm and DMDGP symmetries

In the DMDGP definition, the positions for the first three vertices guarantee that the solution set will contain just incongruent realizations [i.e., the DMDGP solution set does not contain solutions obtained by rotating and translating other solutions (Liberti et al. 2014)] and the strictness of the triangular inequality prevents an uncountable quantity of solutions (Lavor et al. 2012a).

The DMDGP vertex order induces that the search space can be modeled as a graph in a binary-tree shape which will be explored in a depth-first search. In the first three layers of such tree, there is only one node each, as it represents the oneness for the positions of each of the first three vertices. From the fourth vertex on, we have the representations of all the possible positions for the vertices v_i, $i = 4, \ldots, n$, with 2 possibilities for v_4, 4 for v_5, 8 for v_6, and so on (two possible positions for v_i as the intersection of three spheres with centers in one of the possible choices $x_{i-1}, x_{i-2}, x_{i-3} \in \mathbb{R}^3$ for the associated vertices).

By the DMDGP definition and the non-empty solution set assumption, the sphere intersection associated to any three consecutive vertices always results in two possibilities with probability one. The intersection can give just one point when the distance $d_{i-3,i}$, $i = 4, \ldots, n$, is related to torsion angles 0 or π radians (angles defined by the planes given by $v_{i-3}, v_{i-2}, v_{i-1}$ and v_{i-2}, v_{i-1}, v_i), which occurs with probability zero.

In Carvalho et al. (2008), Liberti et al. (2008), a Branch-and-Prune (BP) method is presented for finding all incongruent solutions. The BP algorithm can be exponential in the worst case, which is consistent with the fact that the DMDGP is an NP-hard problem (Lavor et al. 2012a).

Additional distance information related to the pairs $\{v_j, v_i\}$, $j < i - 3$, $i = 5, \ldots, n$, can be used to reduce the search space by pruning infeasible positions in the tree and the search ends when a path from the root of the tree to a leaf node is found by the BP algorithm, such that the positions relative to vertices in the path satisfy the DGP equations (1).

As we already mentioned in the Introduction, DMDGP symmetries are related to partial reflections around planes defined by the Cartesian coordinates of three consecutive vertices. The reflection is “partial” because just part of the structure is reflected, starting from the symmetry vertex, as illustrated in Fig. 1.

The identification of the symmetry vertices is given by the symmetry set (Mucherino et al. 2012), defined by

$$S = \big\{ v \in V : \nexists \{u, w\} \in E \text{ such that } u + 3 < v \leq w \text{ and } v > 3 \big\}, \quad (2)$$

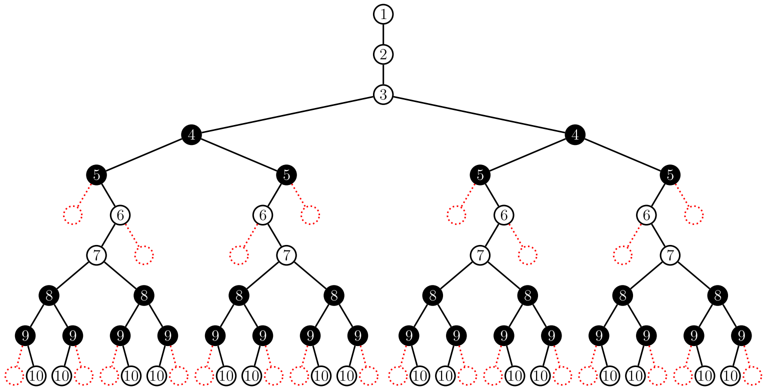

Fig. 2 Symmetric solutions in the BP tree with $S = \{v_4, v_5, v_8, v_9\}$

for a given DMDGP instance $G = (V, E, d)$, where we denote by $u + 3$ the third vertex after u.

The intuition behind the definition of the S is that, if $v \in S$, the partial reflection around the plane defined by the positions of $v-1,\ v-2,\ v-3$ gives another solution, since there is no edge $\{u, w\}$, $u + 3 < v \leq w$, that implies infeasibility of the reflected positions. For example, consider the two possible positions in $\mathbb{R}^3$ for v_4: x_4^0 and x_4^1. Since $v_4 \in S$, for all DMDGP instances, for any solution found in the left subtree of the search space, having the node x_4^0 as its root, there exists another one that is symmetric to the plane defined by the points $x_1,\ x_2,\ x_3 \in \mathbb{R}^3$ (see Fig. 2).

Considering a small DMDGP instance given by

$$
\begin{aligned}
V &= \{v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8, v_9, v_{10}\},\\
E &= \big\{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_4\},\\
&\quad \{v_2, v_3\}, \{v_2, v_4\}, \{v_2, v_5\}, \{v_2, v_6\}, \{v_2, v_7\},\\
&\quad \{v_3, v_4\}, \{v_3, v_5\}, \{v_3, v_6\},\\
&\quad \{v_4, v_5\}, \{v_4, v_6\}, \{v_4, v_7\},\\
&\quad \{v_5, v_6\}, \{v_5, v_7\}, \{v_5, v_8\},\\
&\quad \{v_6, v_7\}, \{v_6, v_8\}, \{v_6, v_9\}, \{v_6, v_{10}\}\\
&\quad \{v_7, v_8\}, \{v_7, v_9\}, \{v_7, v_{10}\}\\
&\quad \{v_8, v_9\}, \{v_8, v_{10}\},\\
&\quad \{v_9, v_{10}\}\big\},
\end{aligned}
$$

we obtain

$$S = \{v_4, v_5, v_8, v_9\}.$$

In Liberti et al. (2013, 2014), it is proved that the cardinality of the DMDGP solution set is $2^{|S|}$ (this occurs with probability 1, for the same reason mentioned in the third paragraph of this Section), which implies that there are 2^4 DMDGP solutions for the example above.

Representing a DMDGP solution by a sequence of zeros and ones, according to the positions obtained by the related sphere intersection, and supposing that the first solution found by BP is given by (remember that the positions for $v_1,\ v_2,\ v_3$ are fixed)

$$s_1 = (0, 0, 1, 0, 0, 0, 1),$$

the other solutions are

$$\begin{array}{lll} s_2 = (0,0,1,0,0,1,0), & s_3 = (0,0,1,0,1,0,1), & s_4 = (0,0,1,0,1,1,0), \\ s_5 = (0,1,0,1,0,0,1), & s_6 = (0,1,0,1,0,1,0), & s_7 = (0,1,0,1,1,0,1), \\ s_8 = (0,1,0,1,1,1,0), & s_9 = (1,0,1,0,0,0,1), & s_{10} = (1,0,1,0,0,1,0), \\ s_{11} = (1,0,1,0,1,0,1), & s_{12} = (1,0,1,0,1,1,0), & s_{13} = (1,1,0,1,0,0,1), \\ s_{14} = (1,1,0,1,0,1,0), & s_{15} = (1,1,0,1,1,0,1), & s_{16} = (1,1,0,1,1,1,0). \end{array}$$

Since there is a symmetry at vertex v_9, the solution s_2 is obtained reflecting the positions for vertices v_9 and v_{10} (around the plane defined by $v_6,\ v_7,\ v_8$) and maintaining the positions for the previous vertices (see Fig. 1). The other solutions are generated using the same idea (see Fig. 2).

In addition to the properties mentioned above, DMDGP symmetries also suggest other ways to explore the DMDGP search space (Fidalgo et al. 2018; Nucci et al. 2013).

3 Finding DMDGP symmetries

The edges of a DMDGP graph $G = (V, E, d)$ can be divided in two disjoint sets,

$$E = E_d \cup E_p,$$

where

$$E_d = \big\{\{v_i, v_j\} \in E : |j - i| \leq 3\big\}$$

is the branching set and

$$E_p = E \backslash E_d$$

is the pruning set (Lavor et al. 2012a).

The branching edges "model" the search space as a binary tree and the pruning edges "indicate" a feasible path. To find a solution, we go down the tree from the root until we reach a leaf node, performing all feasibility tests for the edges in E_p along the way.

From the definition of the set S (2), there is a method to calculate S, in $O(|V||E_p|)$ steps that verifies if there are edges associated with S (Mucherino et al. 2012).

A more efficient method can be defined using the "complement" of S,

$$\overline{S} = \big\{v \in V : \exists\{u, w\} \in E \text{ such that } u + 3 < v \leq w\big\}, \tag{3}$$

implying that

$$S \cup \overline{S} = V \backslash \{v_1, v_2, v_3\} \tag{4}$$

and

$$S \cap \overline{S} = \emptyset.$$

From the definition of $\overline{S}$ (3), we see that only edges in E_p are used to calculate $\overline{S}$. So, let us define, for $i = 1,\ \ldots,\ n - 4$,

$$V_i = \big\{j : \{v_i, v_j\} \in E_p\big\}$$

and

$$m_i = \begin{cases} \max V_i & \text{if } V_i \neq \emptyset, \\ 0 & \text{otherwise,} \end{cases} \tag{5}$$

where $\max V_i = \max\{j : j \in V_i\}$.

Now, we define another set $\mathcal{E}_p$ associated to values $m_i \neq 0$:

1. Set $\mathcal{E}_p := \emptyset$;
2. For $i = 1, \ldots, n-4$, do
   - If $m_i \neq 0$, then $\mathcal{E}_p := \mathcal{E}_p \cup \left\{\{v_i, v_{m_i}\}\right\}$.

Proposition 1 *The set $\overline{S}$ can be rewritten as*

$$\overline{S} = \bigcup_{\{v_i, v_{m_i}\} \in \mathcal{E}_p} \left[v_{i+4}, v_{m_i}\right], \tag{6}$$

where

$$[a, b] = \{v \in V : a \leq v \leq b\}$$

and "$\leq$" is induced by the vertex order.

Proof First, we prove that

$$\overline{S} \subset \bigcup_{\{v_i, v_{m_i}\} \in \mathcal{E}_p} [v_{i+4}, v_{m_i}].$$

If $v_k \in \overline{S}$, there exists $\{v_i, v_j\} \in E_p$ such that $i + 3 < k \leq j$, which implies that $V_i = \left\{j : \{v_i, v_j\} \in E_p\right\}$ is not empty. Since $m_i = \max V_i$, we have $i + 4 \leq k \leq m_i$ and $v_k \in [v_{i+4}, v_{m_i}]$, implying that $v_k \in \bigcup_{\{v_i, v_{m_i}\} \in \mathcal{E}_p} [v_{i+4}, v_{m_i}]$.

Now, let us prove that

$$\bigcup_{\{v_i, v_{m_i}\} \in \mathcal{E}_p} [v_{i+4}, v_{m_i}] \subset \overline{S}.$$

If $v_k \in \bigcup_{\{v_i, v_{m_i}\} \in \mathcal{E}_p} [v_{i+4}, v_{m_i}]$, there exists i such that $\{v_i, v_{m_i}\} \in \mathcal{E}_p$ and $v_k \in [v_{i+4}, v_{m_i}]$. Since $\mathcal{E}_p \subset E_p$, there exists $\{v_i, v_j\} \in E_p$, with $i + 3 < k \leq j$, which means that $v_k \in \overline{S}$. □

To illustrate these new concepts, let us consider the same example given in Section 2. That is,

$$\begin{aligned} E_d = \big\{ & \{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_4\}, \\ & \{v_2, v_3\}, \{v_2, v_4\}, \{v_2, v_5\}, \\ & \{v_3, v_4\}, \{v_3, v_5\}, \{v_3, v_6\}, \\ & \{v_4, v_5\}, \{v_4, v_6\}, \{v_4, v_7\}, \\ & \{v_5, v_6\}, \{v_5, v_7\}, \{v_5, v_8\}, \\ & \{v_6, v_7\}, \{v_6, v_8\}, \{v_6, v_9\}, \\ & \{v_7, v_8\}, \{v_7, v_9\}, \{v_7, v_{10}\}, \\ & \{v_8, v_9\}, \{v_8, v_{10}\}, \\ & \{v_9, v_{10}\}\big\}, \end{aligned}$$

and

$$E_p = \left\{\{v_2, v_6\}, \{v_2, v_7\}, \{v_6, v_{10}\}\right\}.$$

Calculating m_i, for $i = 1, \ldots, 6$, we obtain

$$m_1 = 0, \ m_2 = 7, \ m_3 = 0, \ m_4 = 0, \ m_5 = 0, \ m_6 = 10,$$

which implies that

$$\mathcal{E}_p = \left\{\{v_2, v_7\}, \{v_6, v_{10}\}\right\}$$

and

$$\begin{aligned}\overline{S} &= \bigcup_{\{v_i, v_{m_i}\} \in \mathcal{E}_p} \left[v_{i+4}, v_{m_i}\right], \\ &= [v_6, v_7] \cup [v_{10}, v_{10}] \\ &= \{v_6, v_7, v_{10}\}.\end{aligned}$$

The algorithm for finding the set S is based on the following lemmas.

Lemma 1 *For a given DMDGP instance $G = (V, E, d)$, the set $\mathcal{E}_p$ is calculated in $O(|V| + |E_p|)$ steps.*

Proof Note that we can initialize $m_i = 0$, for $i = 1, \ldots, n-4$, and in one sweep of the edges in E_p, we update the values for m_i. After that, we end up with the final values for m_i and the set $\mathcal{E}_p$ in $O(|V| + |E_p|)$ steps. □

Lemma 2 *For a given DMDGP instance $G = (V, E, d)$ and $\mathcal{E}_p$, the set $\overline{S}$ is calculated in $O(|V|)$ steps.*

Proof This follows from (6) and from the fact that if $\left\{v_i, v_{m_i}\right\}, \left\{v_j, v_{m_j}\right\} \in \mathcal{E}_p$, with $v_i < v_j$ and $v_{j+3} \leq v_{m_i}$, then

$$\left[v_{i+4}, v_{m_i}\right] \cup \left[v_{j+4}, v_{m_j}\right] = \left[v_{i+4}, \max\left\{v_{m_i}, v_{m_j}\right\}\right].$$

□

Lemma 3 *For a given DMDGP instance $G = (V, E, d)$, $\mathcal{E}_p$ and $\overline{S}$, the set S is calculated in $O(|V|)$ steps.*

Proof As the set $\overline{S}$ is ordered, this follows directly from (4). □

A pseudo-code of the method is given in Algorithm 1. Since it is based on the calculation of the sets $\mathcal{E}_p$, $\overline{S}$ and S, in this order, the next result follows directly from the above lemmas.

Theorem 1 *Given a DMDGP instance $G = (V, E, d)$, where $E = E_d \cup E_p$, the associated symmetry set S can be found in $O(|V| + |E_p|)$ steps.*

Corollary 1 *The Algorithm 1 for calculating the symmetry set of a DMDGP instance is optimal.*

Proof From the definition of the set E_d, we obtain that

$$|E| = |E_d| + |E_p| = \Omega(|V| + |E_p|)$$

and, from Theorem 1, we get

$$|E| + |S| = \Omega(|V| + |E_p|) + O(|V| + |E_p|).$$

In order to represent a DMDGP graph $G = (V, E, d)$ and find its symmetry set S, we have at least to read the set E and give as output the set S. Thus, to calculate S, we need $\Omega(|E| + |S|)$ steps, which can be computed in $\Omega(|V| + |E_p|)$ steps. □

Algorithm 1: Optimal way to identify the set S.

Function `OptimalWayToIdentifyS`(V, E_p, K) // $K \in \{2, 3, \ldots\}$
  $\mathcal{E}_p \leftarrow$ `Identify`$\mathcal{E}_p(E_p, |V|, K)$;
  $\overline{S} \leftarrow$ `Identify`$\overline{S}(\mathcal{E}_p, K)$;
  $S \leftarrow \left(V \backslash \{v_1, v_2, v_3, \ldots, v_K\}\right) \backslash \overline{S}$;
  return S;
end

Function `Identify`$\mathcal{E}_p(E_p, n, K)$ // $\{v_i, v_j\} \in E_p$ such that $i < j$
  $M \leftarrow \left\{m_1, m_2, \ldots, m_{n-(K+1)}\right\} \mid m_i = 0\ \forall i$;
  foreach $\{v_i, v_j\} \in E_p$ **do**
    $m_i \leftarrow \max\{m_i, j\}$;
  end
  $\mathcal{E}_p \leftarrow \emptyset$;
  foreach $m_i \in M$ **do**
    if $(m_i \neq 0)$ **then**
      $\mathcal{E}_p \leftarrow \mathcal{E}_p \cup \left\{\{v_i, v_{m_i}\}\right\}$;
    end
  end
  return $\mathcal{E}_p$;
end

Function `Identify`$\overline{S}(\mathcal{E}_p, K)$ // $\{v_i, v_{m_i}\} \in \mathcal{E}_p$ with $i < m_i$ and $\{v_i, v_{m_i}\}, \left\{v_j, v_{m_j}\right\} \in \mathcal{E}_p$ then $v_i < v_j \Leftrightarrow i < j$
  $\overline{S} \leftarrow \emptyset$;
  if $\mathcal{E}_p \neq \emptyset$ **then**
    $i_1 \leftarrow \min\left\{i : \left\{v_i, v_{m_i}\right\} \in \mathcal{E}_p\right\}$;
    $\ell \leftarrow i_1; r \leftarrow m_{i_1}$;
    foreach $\{v_i, v_{m_i}\} \in \mathcal{E}_p$ **do**
      if $\left(r \geq i + K\right)$ **then**
        $r \leftarrow \max\{m_i, r\}$;
      else
        $\overline{S} \leftarrow \overline{S} \cup \left[v_{\ell+(K+1)}, v_r\right]$;
        $\ell \leftarrow i; r \leftarrow m_i$;
      end
    end
    $\overline{S} \leftarrow \overline{S} \cup \left[v_{\ell+(K+1)}, v_r\right]$;
  end
  return $\overline{S}$;
end

4 Final comments

We proposed an optimal algorithm for finding symmetry sets associated with DMDGP instances. We were motivated for distance geometry problems related to molecular geometry ($K = 3$), but note that the dimension K of the realization space is also given as input, which means that the algorithm solves ^{K}DMDGP instances (Liberti et al. 2013).

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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